Jonathan D. Shaw

: jonathan.shaw@colorado.edu
: https://jonathan-d-shaw.github.io

Education

Ph.D. in Applied Mathematics — *Expected May 2028*
University of Colorado Boulder — Boulder, CO
3rd-year Ph.D. student, GPA: 4.0

M.S. in Applied Mathematics — *May 2026*
University of Colorado Boulder — Boulder, CO

B.S. in Applied Mathematics, *magna cum laude* — *May 2023*
University of Colorado Boulder — Boulder, CO
Concentration in Aerospace Engineering

Research Interests

- Numerical PDEs
- Multiscale modeling/simulation
- Particle fluid flows
- Numerical ODEs
- Stiff and/or oscillatory systems
- Transient dynamics in power systems
- Functional, asymptotic, and numerical analysis

Research Experience

Department of Applied Mathematics — University of Colorado Boulder
Research Assistant — *May 2022–Present*

Mobility problem (particulate Stokes flow) — *August 2025–Present*
Project advisor: Eduardo Corona

- Investigated boundary integral methods for elliptic PDE to be applied to the *Mobility problem*, which describes how rigid particles move in highly viscous (Stokes) flows.
- Successfully accelerated the existing implementation of the chosen method.
- Developed a new technique that improves the conditioning of the boundary integral operator.

Fast solvers for power systems applications *May 2023–Present*
Project advisor: Eduardo Corona
Project collaborators: Fiona Majeau, Bri-Mathias Hodge, Amir Sajadi

- Learned fundamental concepts of power system dynamics.
- Studied and implemented various solvers for stiff and/or oscillatory systems of ODEs.
- Wrote code to perform electromagnetic transient simulations using systems and machine parameters from NREL's Sienna package.

Geothermal energy development in Colorado *August 2025–Present*
Project advisors: Bri-Mathias Hodge and *Shae Frydenlund*

- Learned about geothermal energy technologies and conducted a literature review to identify technological challenges to geothermal energy development.
- Designing an ethnographic study aimed at revealing social, economic, and political barriers to geothermal energy development in Colorado.
- Developing a geographic information system (GIS) database that incorporates technological and social barriers to geothermal energy development in Colorado, intended to accelerate local resource development.

Gentrification dynamics *May 2022–September 2024*
Project advisors: Juan Restrepo and *Nancy Rodríguez*

- Developed a dynamical system model of gentrification.
- Performed functional and numerical analysis to understand observed dynamics as a function of model parameters.

Conduit PDE wavepacket interactions *August 2023–September 2023*
Project advisor: Mark Hoefer

- Obtained asymptotic descriptions of small amplitude wavepackets interacting with rarefaction and dispersive shock waves in the context of the conduit PDE.

Los Alamos National Laboratory Los Alamos, NM
Intern - Physics XTD-PRI *May 2018–August 2018*

Algebraic topology for computational fluid dynamics
Project advisor: Roy S. Baty

- Researched fluid dynamics through the scope of algebraic topology.
- Analyzed Schlieren images of supersonic airflow from an elliptic nozzle in order to establish a relationship between the topological characteristics and the time-varying pressure fields of the flow.
- A talk on this research was delivered at the 71st Annual Meeting of the APS Division of Fluid Dynamics (abstract here).

Publications

Peer-Reviewed Articles

1. **Shaw, J. D.**; Restrepo, J. G.; Rodríguez, N., *A dynamical model of gentrification: Exploring a simple rent control strategy*, Phys. Rev. E, 2024 (🌐: arXiv) [3 citations]

Preprints & Other

1. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Analytical Models of Frequency and Voltage in Large-Scale All-Inverter Power Systems*, accepted to IEEE Transactions on Power Systems 2025 (🌐: arXiv) [2 citations]
2. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Computationally Efficient Analytical Models of Frequency and Voltage in Low-Inertia Systems*, accepted to IEEE Transactions on Sustainable Energy 2025 (🌐: arXiv) [0 citations]

Talks & Presentations

Talks

- March 2025, Complex/Dynamical Systems Seminar, University of Colorado Boulder (invited talk)
- January 2025, Dynamics Days US 2025, Denver, CO (contributed talk, travel award recipient)
- October 2023, Mathematical Biology Seminar, University of Colorado Boulder (contributed talk)

Poster Presentations

- January 2025, Dynamics Days US 2025, Denver, CO (contributed poster)
- January 2023, Dynamics Days US 2023, Denver, CO (contributed poster)

Teaching Experience

Department of Applied Mathematics — University of Colorado Boulder
Teaching Assistant — *August 2022–May 2025*

Applied Analysis (APPM 5440/5450) (graduate courses) — *August 2024–May 2025*
Instructor: François G. Meyer

Vector Calculus (APPM 2350) — *May 2024–May 2025*
Instructors: Robert Benim, Eric Thaler

Calculus 1 (APPM 1350) — *August 2023–May 2024*
Instructor: Robert Benim

Discrete Mathematics (APPM 3170) — *August 2022–December 2022*
Instructor: François G. Meyer

Department of Mathematics John Tyler Community College, Richmond, VA
Supplemental Instructor, vector calculus *January 2019–May 2019*

Leadership & Service

Society for Industrial and Applied Mathematics (SIAM) University of Colorado Boulder
President, Graduate Student Chapter *August 2025–Present*

- Organized biweekly applied mathematics graduate and alumni talks, national laboratory tours, and graduate student study sessions.
- Led the chapter in financial planning and budgeting.

Industry Liaison, Graduate Student Chapter *August 2024–May 2025*

- Helped coordinate biweekly applied mathematics graduate and alumni talks.

Buckskin National Youth Leadership Training Richmond, VA
Senior Staff Member *January 2016–August 2023*

- Volunteered with other senior staff members throughout the year to optimize the curriculum and execution of an annual seven-day leadership seminar.
- Gave talks on leadership methodologies and personal leadership development.

Skills

Software: MATLAB, julia, C++, python, LaTeX
Interpersonal: Public speaking, leadership, conflict resolution
Hobbies: Trail running, rock climbing, skiing, reading, cooking

References

Eduardo Corona University of Colorado Boulder
Advisor *Department of Applied Mathematics*
Bri-Mathias Hodge University of Colorado Boulder
Collaborator *Department of Electrical, Computer and Energy Engineering*
Juan Restrepo University of Colorado Boulder
Collaborator *Department of Applied Mathematics*
Nancy Rodríguez University of Colorado Boulder
Collaborator *Department of Applied Mathematics*
François G. Meyer University of Colorado Boulder
Teaching Reference *Department of Applied Mathematics*